



Ridika Naznin

+880-1540-103780 nazninridika2004@gmail.com

linkedin.com/in/ridikanaznin/ github.com/ridika-2004

Education

Islamic University of Technology

Bachelor of Software Engineering

Expected August 2027

Achievements

International Youth Math Challenge (IYMC) 2025: Awarded Silver Medal.

Solved **500+** problems across Competitive Programming Platforms.

Codeforces: 404SkillNotFound — **CodeChef:** ridika2004

Projects

CareerPilot | React.js, Django, MongoDB, ChromaDB | GitHub | Live

- Built a full stack application and RAG pipeline over curated career resources to generate context-aware recommendations, job recommendations, and personalized learning roadmaps.
- Implemented resume parsing, resume-job fit analysis, tailored application generation, and session-based conversational memory using the Gemini API and vector retrieval.

Kernel Memory Allocator for mCertiKOS | C | GitHub

- Extended the mCertiKOS kernel memory manager by implementing buddy allocation, 4MB superpages, and dynamic heap management.
- Reduced memory fragmentation through buddy allocation and efficient page management while maintaining compatibility with the existing operating system architecture.

Comparative Analysis of XLM-RoBERTa and BanglaBERT of Sentiment Analysis | GitHub

- Evaluated XLM-RoBERTa and BanglaBERT for sentiment classification on Bengali and code-mixed datasets using transfer learning.
- Fine-tuned transformer models and evaluated their performance across multiple datasets to analyze the impact of multilingual and monolingual pretraining.

Technical Skills

Languages: C • C++ • Python • Java

Frameworks: Django • FastAPI • Flask

ML: PyTorch • TensorFlow • Transformers

Developer Tools: Visual Studio Code • Git • Docker

Database: MySQL • Oracle • MongoDB

Concepts: Operating System, Data Structures and Algorithms, OOP, Machine Learning, Database Management, Networking, System Design and Architectures

Activities

Co-Instructor — Bangladesh Space Olympiad Season 3

Aug 2025

Conducted the session "Journey Throughout the Time of Universe"

Shaheed Bir Uttam Lt. Anwar Girls' College

Co-Instructor — Summer School'26

May 2026

Conducted the session "Research-Driven Python with Machine Learning Basics"

University of Dhaka

Research Interests

Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing, Multi-Agent Systems.

Current Research: Sentiment Analysis, Named Entity Recognition, Bias Detection